D07 · Cross-Docking & Flow-Through

04 · Merge-in-transit & consolidation hubs

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1 WHAT IT IS

Merge-in-transit consolidates components of one order coming from different origins at a hub, without long-term storage, and ships them together; consolidation hubs more generally combine and redistribute flows between many origins and destinations.

2 WHY IT MATTERS

Customers who order a set expect one delivery, not three. **Merge-in-transit can deliver that experience and convert several partial shipments into one full load**, but every component becomes a single point of failure. It works only when supplier lead times are predictable and advance ship notices are accurate.

3 HOW IT IS DONE

1. **Qualify suppliers on lead-time reliability.** High variability makes synchronization impossible.
2. **Require accurate ASNs with ETAs.** The hub plans the merge from them.
3. **Define the late-component protocol.** Wait or ship partial — decided by customer segment before go-live.
4. **Measure dwell time and assembly accuracy.** Long dwell signals desynchronized suppliers; a missing component is a failed order.

IN PRACTICE

MIT requires all supplier lead times to be predictable and comparable

MIT only works when suppliers have sufficiently predictable lead times for the hub to plan synchronization with precision. Suppliers with high lead time variability are incompatible with MIT.

Advance supplier arrival communication (ASN) is critical for MIT

The MIT hub cannot synchronize 3 supplier arrivals without knowing exactly when each one will arrive. The ASN (Advanced Ship Notice) with each supplier's exact ETA is the prerequisite of efficient synchronization.

Define the management protocol when a component arrives late (single point of failure)

If one of the MIT suppliers doesn't arrive in the agreed window, the hub has two options: wait for the component (delaying customer delivery) or dispatch available components (generating an incomplete delivery). The protocol must be defined before go-live.

MIT has the greatest ROI for orders including items from high-volume multi-supplier

MIT doesn't justify its cost for sporadic or low-volume orders. MIT ROI scales with the volume of multi-supplier orders that can be consolidated at the hub.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Merge-in-transit for white goods dispatch — electronics distributor — A white goods distributor receives orders for complete sets (refrigerator + washer + dryer) from 3 different suppliers.

MIT component	Design	Result
MIT hub in Querétaro (equidistant between 3 suppliers)	4,200 m² hub · No storage racks — only transit zone and docks	Hub lease cost: 40% lower than a standard DC per useful m²
Synchronization of 3 supplier arrivals	Coordinated arrival windows: 8–10am · Complete set consolidated before 11am	98.2% of sets consolidated in <3 hours of last component arrival

MIT component	Design	Result
Complete set dispatch to customer on a single truck	FTL from hub to final customer · Delivery within 24 hours from hub	Customer satisfaction: complete set delivered in one event vs. 3 separate deliveries before MIT

Result: Delivery customer NPS: from 62 (3 separate deliveries) to 84 (complete set delivery). Logistics cost per set: -12% from FTL consolidation vs. 3 supplier LTLs to distributor DCs.

HOW IT IS MEASURED

MIT Assembly Accuracy % (MIT orders completed with all correct components)

MIT Assembly Accuracy %(MIT orders dispatched with all order components correctly consolidated / Total MIT orders) × 100

Benchmark: >99% accuracy in merge-in-transit · A MIT error (missing a component) generates a complete order delivery failure A MIT error resulting in incomplete set delivery to the customer is significantly more costly than a standard picking error — because the customer is already expecting the complete set and an incomplete delivery requires a second delivery event.

MIT Dwell Time (time from first component arrival to complete set dispatch)

MIT Dwell TimeAverage time from first set component arriving to the complete set being dispatched

Benchmark: <4 hours for well-synchronized MIT · >8 hours indicates supplier synchronization problems MIT dwell time measures the effectiveness of supplier synchronization at the hub. A long dwell time means one or more suppliers are arriving significantly before others — generating wait from first to last component.

WHAT YOU WOULD USE

Merge in Transit

SAP EWM — Merge-in-Transit Processing — Module: Multi-Supplier Consolidation + WMS IntegrationSAP EWM supports MIT processing: consolidation hub management, per-order component tracking, and complete set dispatch.

4 SO WHAT

Merge only what you can synchronize. Qualify suppliers on reliability, run on accurate ASNs, decide the late-component rule upfront and measure dwell time relentlessly.

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